

PRESENTATION-AWARE E-VALUES, CROSS-ALLELE PSEUDOSEQUENCE DIFFUSION, AND PROTEOME-SCALE PEPTIDE MATCHING FOR PEPTIDE–MHC ANALYSIS

Mikhail Shugay^{1,2,3*}

¹Institute of Translational Medicine, Russian National Medical State University, Moscow, Russia

²Department of Genomics of Adaptive Immunity, Shemyakin–Ovchinnikov Institute of Bioorganic Chemistry
RAS, Moscow, Russia

³Institute of Molecular Biology NAS RA, Erevan, Armenia

mhcmatch technical appendix

Abstract

This appendix develops the mathematical and statistical theory of `mhcmatch`, the applied peptide–MHC layer built on the `seqtree` fuzzy-search substrate. We take as given the control-calibrated E-value of the `seqtree` appendix (`evaluate.tex`) and specialize it to MHC *presentation*, whose null is allele-conditional and anchor-constrained. We restate the anchor / TCR-facing decomposition and the per-allele presentation-aware E-value (the *forward* restriction problem), then the *reverse* problem—peptide to presenting allele—as a neighbour-vote posterior with a binomial-tail confidence E-value, together with its non-binder filter and class-II promiscuity treatment. The central contribution is a *cross-allele diffusion* model: each allele is represented by its 34-residue groove pseudosequence, and per-anchor preference distributions are smoothed by kernel-weighted empirical-Bayes shrinkage toward groove-similar alleles, with the per-anchor relevance of each groove position *learned* from data (a mutual-information feature importance that separates, e.g., the MHC-I B-pocket governing P2 from the F-pocket governing PΩ). This rescues rare alleles with few peptides—lifting the `seqtree` limitation that distinct alleles are distinct, never-pooled nulls—and we give its hierarchical-Bayes interpretation, effective sample size, limiting cases, and bias–variance trade-off. We then treat proteome-scale scanning (sliding-window presentation with multiple-testing control; near-exact neoantigen source identification), motif logos with length distributions, the composition rules for the planned stability / affinity / cleavage / expression / immunogenicity predictors, and the benchmark protocol against NetMHCpan, NetMHCIIpan, MixMHCpred and MixMHC2pred.

1. SCOPE AND RELATION TO THE SUBSTRATE

`seqtree` provides a payload-agnostic fuzzy-search core and a control-calibrated E-value: for a query q and a scope θ , with an empirical background P_0 estimated from a matched control of size M and

*Correspondence: mikhail.shugay@gmail.com

a target set of size N , the expected number of chance neighbours is $\widehat{E}(q, \theta) = (N/M) n_C(q, \theta)$, with $p_{\text{any}} = 1 - e^{-\widehat{E}}$ and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\widehat{E}) \geq n_D)$ (`seqtree` appendix, Def. of the E-value; Poisson/Chen–Stein bound; Karlin–Altschul [9, 1] as the i.i.d. special case). We do not re-derive that theory; we *specialize* it.

For peptide–MHC the relevant background is not V(D)J generation but *presentation*, which is allele-conditional and constrains the anchor positions. `mhcmatch` owns four developments on top of the substrate: (i) the productionized forward/reverse presentation E-value (§2–3); (ii) the cross-allele *pseudosequence diffusion* model that pools statistics across groove-similar alleles (§4, the headline); (iii) proteome-scale scanning and near-exact source identification (§5); and (iv) motif logos (§6). Planned predictors and the comparison protocol are §7–8; limitations are §9.

2. ANCHOR DECOMPOSITION AND THE PER-ALLELE PRESENTATION E-VALUE

A presented peptide of length L factorizes $\sigma = (\sigma_A, \sigma_T)$ into MHC-facing *anchors* $A(a, L)$ and TCR-facing positions T . For class I the buried anchors are {P2, PΩ} (the B- and F-pockets); for the class II 9-mer core they are {P1, P4, P6, P9}. *Presentation* fixes σ_A ; *recognition* reads σ_T . Dolton et al. [6] exhibit one HLA-A02:01 TCR cross-reacting with three epitopes through a shared central motif with the anchors irrelevant, which motivates two complementary maskings, each realized by a per-position weight vector w over a `seqtree PositionalMatrix`:

$$\text{(recognition)} \quad w_i = \mathbb{1}[i \in T]; \quad \text{(presentation)} \quad w_i = \mathbb{1}[i \in A(a, L)]. \quad (1)$$

In `mhcmatch` these are exposed by `Store.decompose`, which returns both readouts with the masked positions written as the spare symbol `X`.

FORWARD PROBLEM.. For a candidate restriction allele a with an anchor-masked presented control C_a of size M_a and target peptidome of size N_a , the presentation-aware E-value is the substrate E-value on allele-restricted, anchor-conditioned indices,

$$\widehat{E}(q, \theta; a) = \frac{N_a}{M_a} n_{C_a}(\sigma_T, \theta), \quad p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\widehat{E}) \geq n_{D_a}^T), \quad (2)$$

i.e. significance is computed on the TCR-facing readout only, so shared anchors (presentation, not recognition) do not inflate within-allele hits. The analytic null for rare a is taken up in §4.

3. REVERSE PROBLEM: PEPTIDE TO PRESENTING ALLELE

The forward E-value answers “is q a binder of a *given* a ?”. Restriction prediction asks the reverse: which alleles present q ? We flip the weights in (1) to the *presentation* mode and index reference peptides by their anchor signature (`layout.presentation_features`; class I N/C-pocket residues P1 P2 P3, P(Ω–1), PΩ, class II core P1 P4 P6 P9). For a query we widen the scope on its signature until it has $n \in [10, 100]$ non-exact neighbours, and tally their alleles, k_a votes out of n .

DEFINITION 1 (Ranking and confidence). The *ranking* score is the neighbour vote fraction $\hat{p}(a \mid q) = k_a/n$, a posterior over the panel. The *confidence* score is the per-allele enrichment over the panel

background frequency f_a ,

$$E_a = -\log_{10} \mathbb{P}(\text{Binomial}(n, f_a) \geq k_a), \quad (3)$$

the upper binomial tail; a peptide binds a iff $E_a \geq -\log_{10} \alpha$ and $k_a > 0$.

REMARK 1 (Why vote fraction, not enrichment, ranks). On a skewed panel the dominant true allele has a large expected count nf_a , so the enrichment (3) *penalises* it; the vote fraction does not. Hence $\hat{p}$ ranks and E_a gates. This separation, validated on `pmhc_data` (per-(peptide, allele) ROC-AUC 0.90–0.99, `seqtree/bench/bench_mhc_guess.py`), is what `Store.restriction` implements.

CLASS-II REGISTER TRICK.. The class-II core register inside a longer peptide is unknown; full deconvolution clusters it [3, 2]. We use a one-pass proxy: pick the single 9-mer window whose P1 is a large hydrophobic and whose P4/P6/P9 avoid Pro/Gly, and take that core’s {P1,P4,P6,P9}. Committing to one register makes an allele’s peptides share a consistent signature and is what lifts class-II ROC-AUC into the 0.9 range.

NON-BINDER FILTER AND PROMISCUITY.. A peptide that binds *no* panel allele has small $\max_a E_a$ (real-vs-random rejection); one that does not bind a *specific* a has $E_a < -\log_{10} \alpha$. The global statistic $E_{\text{glob}} = \sum_a 10^{-E_a}$ is the expected number of panel alleles presenting q by chance. Class II is genuinely multi-label (an open groove with loosely specified pockets presents one peptide on many alleles), so the panel positive set is *every* observed restricting allele and the non-binder test is “binds none of the panel”.

4. MHC PSEUDOSEQUENCE MODEL AND CROSS-ALLELE DIFFUSION

`seqtree` deliberately treats distinct alleles as distinct, never-pooled nulls. This is correct but wasteful: rare alleles carry few presented peptides, so both (2) and (3) are noisy exactly where data are thin—the equity problem of Glynn et al. [8]. `mhcmatch` lifts the limitation by making the allele a *sequence* the same engine can compare, and diffusing statistics along groove similarity.

PSEUDOSEQUENCE.. Each allele a is represented by its NetMHCpan-style 34-residue groove pseudosequence $s_a \in \Sigma^{34}$ [14]: the polymorphic, peptide-contacting positions (class I α_1/α_2 ; class II $\alpha_1 + \beta_1$). Per-allele binding can be predicted from s_a alone [8, 4], so groove distance is a meaningful allele distance.

4.1. ANCHOR-FACTORED SIMILARITY KERNEL

The presentation motif factorizes by pocket: anchor j ’s residue preference is governed by the groove positions lining pocket j , and different pockets are spatially separated (the class-I B-pocket sets P2, the F-pocket sets PΩ). We therefore use a *per-anchor* kernel rather than one global allele distance. For anchor j let $w_j \in \mathbb{R}_{\geq 0}^{34}$ weight the groove positions, and define

$$d_j(a, b) = \sum_{p=1}^{34} w_{j,p} \mathbb{1}[s_{a,p} \neq s_{b,p}], \quad K_j(a, b) = \exp(-d_j(a, b)/h_j), \quad (4)$$

a weighted-Hamming kernel with bandwidth $h_j > 0$ (ambiguous positions X are skipped). Setting $w_j \equiv 1$ recovers a single un-factored kernel.

DEFINITION 2 (Learned per-anchor relevance). Let A_j be the allele’s modal residue at peptide anchor j (from its presented set) and S_p the groove residue at pseudosequence position p , both viewed as categorical variables over the allele panel. The relevance of position p to anchor j is the mutual information

$$w_{j,p} \propto I(S_p; A_j) = \sum_{x,y} \hat{\mathbb{P}}(S_p=x, A_j=y) \log_2 \frac{\hat{\mathbb{P}}(S_p=x, A_j=y)}{\hat{\mathbb{P}}(S_p=x) \hat{\mathbb{P}}(A_j=y)}, \quad (5)$$

normalized to mean one over p . Equivalently w_j may be fit by maximizing the held-out predictive log-likelihood of the shrinkage model below under an ℓ_1 /entropy penalty, which selects each anchor’s pocket positions. (5) is the “feature importance” that says which groove residues drive P2 versus PΩ.

4.2. KERNEL-SHRINKAGE POOLING

Write $\theta_{a,j}$ for allele a ’s empirical residue distribution at anchor j over its n_a peptides. We shrink it toward groove-similar alleles:

$$\hat{\theta}_{a,j} = \frac{n_a \theta_{a,j} + \sum_{b \neq a} K_j(a, b) n_b \theta_{b,j}}{n_a + \sum_{b \neq a} K_j(a, b) n_b}, \quad \text{ESS}_{a,j} = n_a + \sum_{b \neq a} K_j(a, b) n_b. \quad (6)$$

PROPOSITION 1 (Hierarchical-Bayes reading and limits). (6) is the posterior mean of a Dirichlet–multinomial in which the neighbour-weighted counts form the prior pseudocounts: $\hat{\theta}_{a,j} = \mathbb{E}[\theta | \text{counts}]$ with concentration $\text{ESS}_{a,j}$. As $h_j \rightarrow 0$, $K_j(a, b) \rightarrow 0$ for $b \neq a$ and $\hat{\theta}_{a,j} \rightarrow \theta_{a,j}$ (no pooling); as $h_j \rightarrow \infty$, $K_j(a, b) \rightarrow 1$ and $\hat{\theta}_{a,j} \rightarrow (\sum_b n_b \theta_{b,j}) / \sum_b n_b$ (the global pool).

(Both limits are implemented and unit-tested in `Pseudoseq.shrink`.)

PROPOSITION 2 (Bias–variance and bandwidth). Treat each $\theta_{b,j}$ as an unbiased estimate of allele b ’s truth $\theta_{b,j}^*$ with variance $\propto 1/n_b$. The pooled estimator has variance $O(1/\text{ESS}_{a,j})$ and bias $\|\sum_b \kappa_b (\theta_{b,j}^* - \hat{\theta}_{a,j})\|$, where $\kappa_b = K_j(a, b) n_b / \text{ESS}_{a,j}$ is the borrowed mass. The mean-squared error is minimized by borrowing more (smaller effective h_j penalty) when n_a is small and groove neighbours are close, and less when n_a is large; calibrating h_j by cross-validated predictive likelihood realizes this trade-off automatically.

POOLED NULL AND RESTORED E-VALUE.. For a rare allele the analytic presented null is built from the shrunk anchor marginals: assuming approximate position independence in the anchor block (the maximum-entropy presented distribution of [15] conditioned on the motif), $\widehat{P}_0(\sigma_A | a) \approx \prod_j \hat{\theta}_{a,j}(\sigma_{A_j})$, which feeds a usable $\widehat{E}$ in (2)–(3) where the raw per-allele control was too small. The estimator is a smoothed null: its bias is controlled by the borrowed mass on dissimilar alleles (Prop. 2), vanishing as the panel around a becomes dense and groove-tight.

4.3. CLUSTERING AND PROMISCUITY

Thresholding K_j (or the un-factored K) yields single-linkage allele *clusters* of functionally similar alleles (`Pseudoseq.cluster`). A peptide presented across a cluster is *promiscuous*; its promiscuity is the spread of its positive alleles over the clustering, and the cluster is the natural unit over which to pool nulls when per-allele data are thin (especially class II). This is the principled cross-allele similarity that the substrate omits.

5. PROTEOME-SCALE MATCHING

PRESENTATION SCAN.. To find all presented peptides in a protein we slide every binding-length window (class I 8–11; class II ≥ 13 with the register-anchored core) and evaluate (3) per window and allele. With W windows and $|\mathcal{A}|$ panel alleles the test is multiple: under the null the expected number of windows called for allele a is $\approx W\alpha$, so we control the family either by Bonferroni / max-statistic thresholds (FWER) or by Benjamini–Hochberg on the p_{enrich} across the $W \times |\mathcal{A}|$ grid (FDR) [5]. `scan_protein` reports per-window binders; the grid-level correction is a calibration step.

SAME-MHC VERSUS TCR-FACING SEARCH.. At proteome scale two notions of “similar peptide” are distinguished by which mask of (1) is indexed: *same-MHC* similarity indexes the presentation signature (peptides likely presented by the same allele), *TCR-facing* similarity indexes the anchor-masked central motif (similar T-cell recognition, the basis of cross-reactivity). Both run on the C++ KmerIndex seed-and-gather.

MOLECULAR MIMICRY.. A self or foreign peptide p is a candidate cross-reactive mimic of a neoantigen ν iff (i) it is presented by a compatible allele, (ii) the TCR-facing distance $d_{\text{TCR}}(p, \nu) = \sum_i w_i \text{pen}(p_i, \nu_i) \leq \theta$ (anchors zeroed), and (iii) the hit is significant against the per-allele presented background, $\hat{E} \leq \alpha$. Condition (iii) deflates motifs common in the allele’s peptidome and elevates surprisingly shared ones. This connects to the neoantigen-quality program of Łuksza et al.: the quality $Q = R \times D$ [11, 10] multiplies a recognition / non-self term R (the cross-reactivity distance to known antigens) by a self-discrimination term D (differential MHC binding), and to the close-to-self “shell” picture of [15]; `mhcmatch` supplies a calibrated E-value for the R component.

NEAR-EXACT SOURCE IDENTIFICATION.. Given a neoantigen we identify the self peptide it derives from—its parent protein and the mutated position—by full-sequence (unmasked) $\leq m$ -mismatch search over all length- L windows of the reference proteome. Under an i.i.d. background of composition $\{q_c\}$ the expected number of chance windows within m substitutions of a query in a proteome of R residues is

$$\mathbb{E}[\#\text{chance}] \approx R \sum_{i=0}^m \binom{L}{i} \bar{q}^{L-i} (1 - \bar{q})^i, \quad \bar{q} = \sum_c q_c^2, \quad (7)$$

the Karlin–Altschul/birthday count specialized to the Hamming ball (the empirical-background variant reuses `seqtree` §KA). For $m \in \{0, 1\}$ and $L \geq 8$, (7) is $\ll 1$, so a hit is essentially certainly the true source rather than coincidence—`Proteome.find_source` returns it with the differing residue flagged.

6. MOTIF LOGOS AND LENGTH DISTRIBUTIONS

For an allele’s presented set (class I peptides of a fixed length; class II register-anchored 9-mer cores) the per-position information content is

$$I_i = \log_2 20 - H_i - e(n_i), \quad H_i = - \sum_c f_{i,c} \log_2 f_{i,c}, \quad (8)$$

with $f_{i,c}$ the residue frequency at position i and $e(n_i)$ the small-sample entropy correction for n_i counts (negligible for the deep `pmhc_data` alleles). The logo draws letter c at height $f_{i,c} I_i$ bits; columns at

the anchors stand tall (e.g. A02:01 shows P2=L, PΩ ∈ {V,L}), the TCR-facing columns are flat. The length distribution is reported as a companion histogram. Columns may be confidence-weighted by n (or by the shortlist ≥ 2 -publication support). `logo.motif` returns the numeric logo; `logo.render` draws it.

7. PLANNED PREDICTORS AND THEIR COMPOSITION

Presentation is necessary but not sufficient for immunogenicity. Each planned predictor scores a distinct biophysical or population term and *composes* with the presentation evidence on the log scale; for peptide σ and allele a the combined log-odds is

$$\log \frac{\mathbb{P}(\text{immunogenic})}{\mathbb{P}(\text{not})} = \beta_0 + \beta_{\text{pres}} (-\log p_{\text{enrich}}) + \beta_{\text{aff}} \log \frac{1}{\text{IC}_{50}} + \beta_{\text{stab}} t_{1/2} + \beta_{\text{clv}} c_{\text{Cterm}} + \beta_{\text{expr}} \log e_{\text{gene}} + \beta_{\text{imm}} \rho_{\text{TCR}}, \quad (9)$$

with coefficients fit on labelled outcome data (the user will supply tuning/benchmark sets). The terms: *affinity* IC_{50} and *stability* $t_{1/2}$ are the quantitative complements of the binary presentation E-value, regressable on the same anchor features plus the pseudosequence; *cleavage* c_{Cterm} is a position-specific C-terminal generation probability (with optional N-terminal trimming); *expression* e_{gene} and variant frequency enter as Bayesian priors multiplying the presentation odds; *immunogenicity* ρ_{TCR} combines physicochemical TCR-facing features with a TCR precursor-frequency / generation-probability estimate [12, 7] in the $Q = R \times D$ spirit of [11]. The precursor-frequency estimator may live in a separate package and be consumed here. Each term is a roadmap Phase-2 milestone specified by this subsection.

8. BENCHMARK AND EVALUATION METHODOLOGY

We evaluate as a per-(peptide, allele) binary task. *Splits* are by allele *and* by peptide: held-out alleles test cross-allele generalization (and the diffusion model of §4), held-out peptides within an allele test within-allele recall; clonotype/ peptide duplicates are collapsed before splitting to prevent leakage. *Metrics* are ROC-AUC, PR-AUC against the prevalence baseline, top-1 accuracy, and a real-vs-random noise-rejection AUROC (the non-binder filter). *Comparison*: against NetMHCpan-4.1 and NetMHCIIpan-4.0 [14], MixMHCpred [4], and MixMHC2pred [13] on matched alleles and shared eluted-ligand / assay test sets, with rank-threshold calibration and an explicit *rare-allele subgroup* analysis (the equity axis of [8]) where the pseudosequence diffusion is expected to help most. The benchmark datasets and the paper live in a dedicated repository; this section fixes the protocol so it stays consistent with the predictors defined here.

9. ASSUMPTIONS AND LIMITATIONS

(i) The analytic pooled null assumes approximate position independence in the anchor block; real co-occupancy adds a two-point (b_2) correction (the negative-binomial tail of the `seqtree` appendix) that keeps the mean robust while inflating the tail. (ii) Kernel validity rests on a faithful pseudosequence; ambiguous (X) positions are skipped and locus-aware normalization of allele names is required

(class II especially). (iii) The pooled E-value is a smoothed estimator (Prop. 2)—its bias must be reported alongside the gain for rare alleles. (iv) The class-II register trick is a one-pass proxy for full deconvolution [3, 2]. (v) “Compatible allele” in mimicry is a user input; cross-reactivity significance is conditional on a faithful per-allele presented control. These are the open items tracked in ROADMAP.md.

REFERENCES

- [1] Stephen F. Altschul, Warren Gish, Webb Miller, Eugene W. Myers, and David J. Lipman. Basic local alignment search tool. *Journal of Molecular Biology*, 215(3):403–410, 1990.
- [2] Bruno Alvarez, Birkir Reynisson, Carolina Barra, Søren Buus, Nicola Ternette, Tim Connelley, Massimo Andreatta, and Morten Nielsen. NAlign_MA; MHC peptidome deconvolution for accurate MHC binding motif characterization and improved T-cell epitope predictions. *Molecular & Cellular Proteomics*, 18(12):2459–2477, 2019.
- [3] Massimo Andreatta, Bruno Alvarez, and Morten Nielsen. GibbsCluster: unsupervised clustering and alignment of peptide sequences. *Nucleic Acids Research*, 45(W1):W458–W463, 2017.
- [4] Michal Bassani-Sternberg, Chloé Chong, Philippe Guillaume, Marthe Solleder, HuiSong Pak, Philippe O. Gannon, Lana E. Kandalaft, George Coukos, and David Gfeller. Deciphering HLA-I motifs across HLA peptidomes improves neo-antigen predictions and identifies allosteric regulating HLA specificity. *PLOS Computational Biology*, 13(8):e1005725, 2017.
- [5] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.
- [6] Garry Dolton, Cristina Rius, Aaron Wall, et al. Targeting of multiple tumor-associated antigens by individual T cell receptors during successful cancer immunotherapy. *Cell*, 186(16):3333–3349, 2023.
- [7] Yuval Elhanati, Anand Murugan, Curtis G. Callan, Thierry Mora, and Aleksandra M. Walczak. Quantifying selection in immune receptor repertoires. *Proceedings of the National Academy of Sciences USA*, 111(27):9875–9880, 2014.
- [8] Eric Glynn, Dario Gherzi, and Mona Singh. Toward equitable major histocompatibility complex binding predictions. *Proceedings of the National Academy of Sciences USA*, 122(8):e2405106122, 2025.
- [9] Samuel Karlin and Stephen F. Altschul. Methods for assessing the statistical significance of molecular sequence features by using general scoring schemes. *Proceedings of the National Academy of Sciences USA*, 87(6):2264–2268, 1990.
- [10] Marta Łuksza, Nadeem Riaz, Vladimir Makarov, Vinod P. Balachandran, Matthew D. Hellmann, Alexander Solovyov, Naiyer A. Rizvi, Taha Merghoub, Arnold J. Levine, Timothy A. Chan,

- Jedd D. Wolchok, and Benjamin D. Greenbaum. A neoantigen fitness model predicts tumour response to checkpoint blockade immunotherapy. *Nature*, 551(7681):517–520, 2017.
- [11] Marta Łuksza, Zachary M. Sethna, Luis A. Rojas, Jayon Lihm, Barbara Bravi, Yuval Elhanati, Kevin Soares, Masataka Amisaki, Anton Dobrin, David Hoyos, Pablo Guasp, Abderezak Zeboudj, Rebecca Yu, Adrienne Kaya Chandra, Theresa Waters, Zagaa Odgerel, Joanne Leung, Rajya Kappagantula, Alvin Makohon-Moore, Amber Johns, Anthony Gill, Mathieu Gigoux, Jedd Wolchok, Taha Merghoub, Michel Sadelain, Erin Patterson, Remi Monasson, Thierry Mora, Aleksandra M. Walczak, Simona Cocco, Christine Iacobuzio-Donahue, Benjamin D. Greenbaum, and Vinod P. Balachandran. Neoantigen quality predicts immunoediting in survivors of pancreatic cancer. *Nature*, 606(7913):389–395, 2022.
- [12] Anand Murugan, Thierry Mora, Aleksandra M. Walczak, and Curtis G. Callan. Statistical inference of the generation probability of T-cell receptors from sequence repertoires. *Proceedings of the National Academy of Sciences USA*, 109(40):16161–16166, 2012.
- [13] Julien Racle, Justine Michaux, Georg Alexander Rockinger, Marion Arnaud, Sara Bobisse, Chloe Chong, Philippe Guillaume, George Coukos, Alexandre Harari, Camilla Jandus, Michal Bassani-Sternberg, and David Gfeller. Robust prediction of HLA class II epitopes by deep motif deconvolution of immunopeptidomes. *Nature Biotechnology*, 37(11):1283–1286, 2019.
- [14] Birkir Reynisson, Bruno Alvarez, Sinu Paul, Bjoern Peters, and Morten Nielsen. NetMHCpan-4.1 and NetMHCIIpan-4.0: improved predictions of MHC antigen presentation by concurrent motif deconvolution and integration of MS MHC eluted ligand data. *Nucleic Acids Research*, 48(W1):W449–W454, 2020.
- [15] Andreas Tiffeau-Mayer, Jonathan A. Levine, Christopher J. Russo, Quentin Marcou, William Bialek, and Benjamin D. Greenbaum. How different are self and nonself? *PRX Life*, 4:013027, 2026.